ALOHA, Slotted ALOHA & Collision-Free Protocols

Computer Networks — Multiple Access Protocols

Introduction

In a network, multiple computers often share the same communication channel. If two or more computers transmit data simultaneously, their signals may overlap and become corrupted — this is called a **collision**. Multiple Access Protocols control how devices share a common channel and reduce or eliminate collisions.

1. Pure ALOHA

Definition

Pure ALOHA is the **simplest random access protocol** in which a station transmits data whenever it has data to send — without checking whether the channel is free or busy. If two stations transmit at the same time, a collision occurs and frames are destroyed. Stations then wait for a **random time** before retransmitting.

How It Works

- A station generates a frame.
- It **immediately transmits** the frame.
- If an ACK is received → transmission is successful.
- If no ACK is received → collision is assumed.
- The station waits for a **random time** and retransmits.

Example

Computer A ■■■ Frame A   Computer B ■■■ Frame B
Since both transmissions overlap, a **collision occurs** and both frames are lost. Both computers then wait for random intervals before retrying.

Real-Life Analogy

■■ Like a **walkie-talkie**: anyone can speak at any time. If two people speak simultaneously, neither message is understood — both must repeat.

Vulnerable Time & Efficiency

Vulnerable Time = 2 × Frame Transmission Time (2T)
Maximum Efficiency = 18.4% — most bandwidth is wasted due to collisions.

Advantages & Disadvantages

Advantages	Disadvantages
✔ Very simple to implement	✘ High collision probability
✔ No synchronization required	✘ Poor channel utilization

✔ Easy to understand	✘ Low efficiency (18.4%)

2. Slotted ALOHA

Definition

Slotted ALOHA is an **improved version of Pure ALOHA**. Time is divided into equal-sized slots, and a station can transmit **only at the beginning of a slot**. This restriction halves the probability of collision.

How It Works

- Time is divided into **equal slots**.
- A station waits until the **beginning of a slot**.
- It transmits only when a slot starts.
- If two stations choose the **same slot**, a collision occurs.
- After collision, stations wait for **random slots** before retransmitting.

Example

```
| Slot 1 | Slot 2 | Slot 3 | Slot 4 |
If Station A transmits in Slot 2 and no other station uses Slot 2 → success.
If both A and B transmit in Slot 2 → collision.
```

Real-Life Analogy

■ Like a **classroom** where students can speak only when the teacher says "Start." Waiting for fixed opportunities reduces random interruptions.

Vulnerable Time & Efficiency

Vulnerable Time = T (half of Pure ALOHA!)
Maximum Efficiency = 36.8% — nearly **twice** the efficiency of Pure ALOHA.

Advantages & Disadvantages

Advantages	Disadvantages
✔ Fewer collisions	✘ Requires synchronization
✔ Better bandwidth utilization	✘ More complex than Pure ALOHA
✔ Higher efficiency (36.8%)	

Comparison: Pure ALOHA vs Slotted ALOHA

Feature	Pure ALOHA	Slotted ALOHA
Transmission	Anytime	Only at slot boundaries
Synchronization	Not Required	Required
Vulnerable Time	2T	T
Efficiency	18.4%	36.8%
Collision Probability	High	Lower

Performance	Poor	Better

3. Collision-Free Protocols

Definition

Collision-Free Protocols are channel access methods designed to **completely eliminate collisions**. They decide in advance which station is allowed to transmit — unlike ALOHA, where devices transmit first and deal with collisions afterward.

Why Are They Needed?

In networks with **heavy traffic**, collisions waste bandwidth. Collision-Free Protocols provide organized channel access, resulting in reliable communication and better utilization of network resources.

3a. Bit-Map Protocol

Every station is assigned a bit position. Stations that want to transmit set their corresponding bit to **1**. After collecting the bit map, stations transmit one by one in a predefined order.

Example

```
4 stations: Station 0, Station 1, Station 2, Station 3
Stations 0 and 2 want to transmit → Bit Map = 1010
Transmission order: Station 0 → Station 2
No collision because the order is predetermined.
```

Real-Life Analogy

■ Like students **raising their hands** in class. The teacher calls them one at a time — students speak in sequence, so interruptions are avoided.

Advantages	Disadvantages
✔ No collisions	✘ Additional overhead
✔ Fair access	✘ Less efficient under light traffic
✔ Efficient under heavy traffic	

3b. Token Passing Protocol

Token Passing uses a special control frame called a **token**. The token continuously moves among stations. Only the station **currently holding the token** can transmit data.

How It Works

```
Stations: A → B → C → D → A (circular)
When a station receives the token:
   • Has data → transmits
   • No data → passes token to next station
```

Real-Life Analogy

■ Like a meeting where only the person **holding the microphone** can speak. After finishing, the microphone is passed to the next person.

Advantages	Disadvantages
✔ No collisions	✘ Token management required
✔ Fair channel access	✘ If token is lost, comm stops
✔ Suitable for heavy traffic	✘ More complex

ALOHA vs Collision-Free Protocols

Feature	ALOHA	Collision-Free
Collision	Possible	Not Possible
Retransmission	Required	Rarely Required
Efficiency	Lower	Higher
Complexity	Simple	More Complex
Traffic Handling	Better for Light Traffic	Better for Heavy Traffic

Relationship with OS Concepts

Collision-Free Protocols are very similar to **synchronization mechanisms in Operating Systems**:

Operating System	Networking
Critical Section	Shared Channel
Process	Network Station
Semaphore / Mutex	Token
Synchronization	Collision Control

Just as only **one process can enter a critical section** at a time, only **one station can transmit** on the channel at a time in Collision-Free Protocols.

Quick Revision

Pure ALOHA	Slotted ALOHA	Collision-Free
• Send anytime • No sync • Vuln Time = 2T • Efficiency = 18.4% • High collisions	• Send at slot start • Sync required • Vuln Time = T • Efficiency = 36.8% • Fewer collisions	• No collisions • Bit-Map & Token • Heavy traffic use • Higher efficiency

■ Memory Tricks

Pure ALOHA: "Send whenever you want."

Slotted ALOHA: "Send only when the slot starts."

Collision-Free: "Take permission first, then send."